

Lab 3: Architecture Selection & Justification

Internal_Microservice_Catalog_Health_Portal

Architecture Selection

We chose Microservices Architecture for the Internal Microservice Catalog & Health Portal in the Developer Tools & IT Operations domain.

Reason 1

The Service Catalog, Health Monitoring, Authentication, and Alerting functions can be developed, deployed, and scaled independently. This is useful for DevOps Engineers and System Architects who manage many internal services.

Reason 2

Microservices provide fault isolation. If the alerting or health-monitoring service becomes unavailable, users can still access the catalog and view stored service information.

Security Advantage

The Authentication & Authorization component verifies users and restricts administrative actions, such as registering services or changing ownership and endpoint information, to authorized personnel.

Performance Benefit

The portal can retrieve catalog metadata and health information through separate APIs. Each service can be scaled according to demand, improving response time when many developers or architects access the portal.

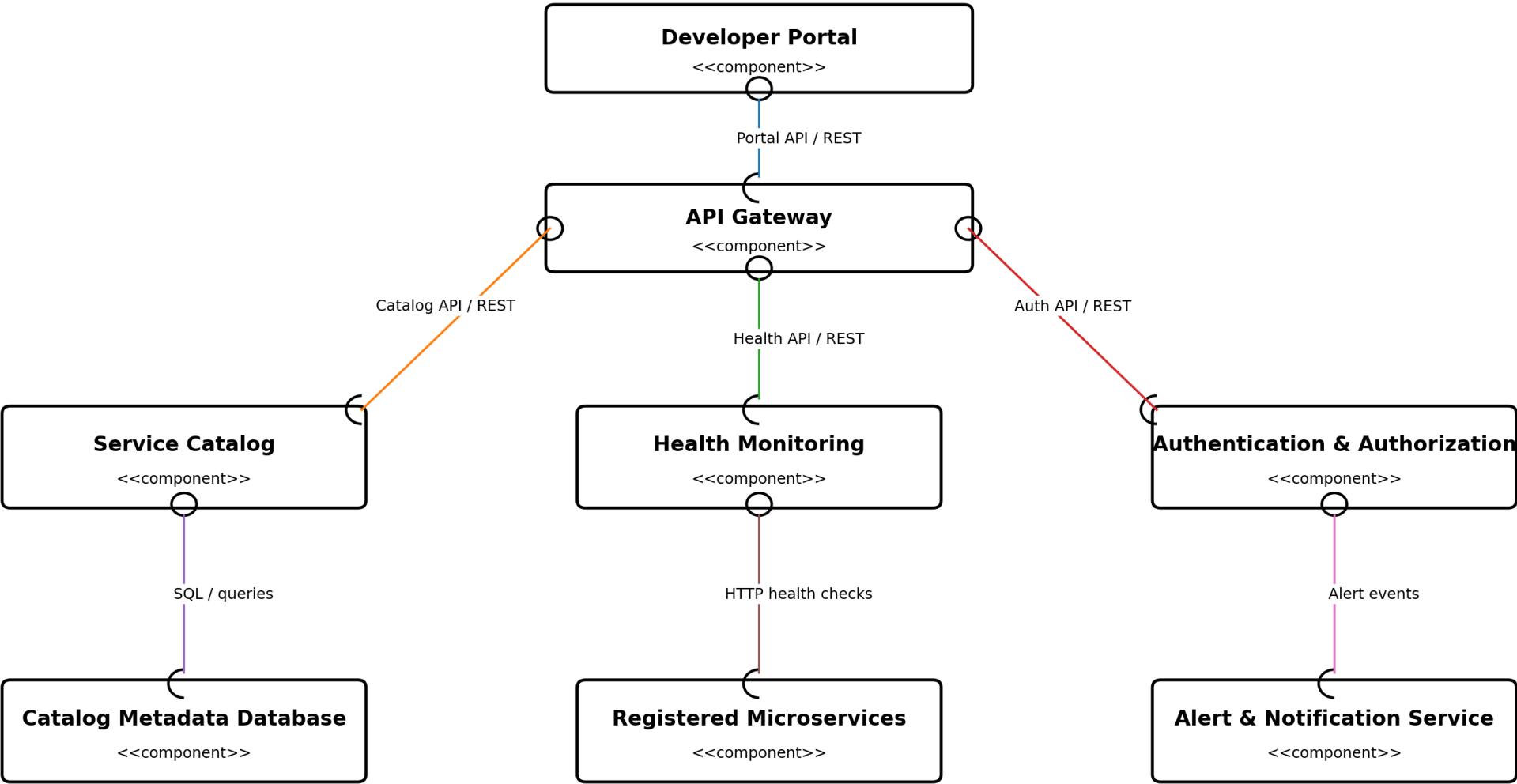
Main Components

Developer Portal, API Gateway, Service Catalog, Health Monitoring, Authentication & Authorization, Catalog Metadata Database, Registered Microservices, and Alert & Notification Service.

Internal Microservice Catalog & Health Portal — Component Diagram

Internal Microservice Catalog & Health Portal

Developer Tools & IT Operations | Users: DevOps Engineer, System Architect



Legend ○ Provided interface (ball) ⌋ Required interface (socket)